

Recommender Module – Task Breakdown

Phase 1: Data Preparation

1. Define Property Data Schema

- Decide required fields:
 - Property ID
 - Price
 - Location
 - Bedrooms
 - Bathrooms
 - Area (sqm)
 - Property type
 - Amenities
 - Date listed
 - Popularity score (simulated)

2. Collect or Generate Dataset

- Use:
 - Mock dataset (CSV / DB)
 - Or generate synthetic data if needed
- Ensure data is clean and consistent.

3. Data Cleaning & Preprocessing

- Handle missing values
- Normalize numerical features (price, area)
- Encode categorical features:
 - Location

- Property type
 - Convert amenities into vector form (multi-hot encoding)
-

Phase 2: Content-Based Engine

4. Define User Preference Input Structure

Create a structured input format

5. Feature Vectorization

- Convert property features into numerical vectors
- Convert user preferences into same vector space

6. Similarity Calculation

Choose one approach:

- Cosine similarity
OR
- Weighted scoring system

Example scoring:

- Location match → weight 0.3
- Price match → weight 0.25
- Bedrooms → weight 0.2
- Amenities → weight 0.15
- Property type → weight 0.1

7. Implement Content Match Score

Return similarity score between user preference and each property.

Phase 3: Hybrid Enhancement

8. Add Popularity Score

- Simulate views/favorites
- Normalize popularity score between 0–1

9. Add Recency Score

- Calculate days since listing
- Newer listings get higher score

10. Define Final Ranking Formula

Example:

Final Score =

$0.7 \times \text{ContentMatch}$

- $0.2 \times \text{Popularity}$
- $0.1 \times \text{Recency}$

Make weights configurable.

Phase 4: Ranking & Output

11. Sort Properties

- Sort by Final Score descending.

12. Select Top N Results

- Return top 5 or top 10.

13. Format JSON Response

Return structured output

Phase 5: API Development

14. Create FastAPI Endpoint

Example:

POST /recommend

15. Input Validation

- Validate user preference schema
- Handle missing fields gracefully

16. Error Handling

- No matching properties
 - Invalid values
 - Empty dataset
-

Phase 6: Integration Preparation

17. Connect with Chatbot Module

- Ensure recommender accepts structured JSON from chatbot.

18. Test with Sample Inputs

- Budget too low
- Rare amenities
- Only location given
- Full preference given

19. Performance Optimization (Optional)

- Precompute feature matrix
 - Avoid recomputing vectors each request
-

Phase 7: Documentation

20. Document:

- Scoring logic
 - Weight justification
 - Limitations (no real collaborative data)
 - Future improvement plan
-

Deliverables of Recommender Module

By the end, you should have:

- Clean dataset
- Content similarity engine
- Hybrid ranking formula
- REST API endpoint
- Tested integration-ready module
- Documented architecture